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To the Editors
MethodsX

Dear Editors,

We are pleased to submit our manuscript entitled “A Comparative Analysis of Open-Source LLMs for Converting Architecture, Engineering and Construction Standards to the RASE Format” for consideration in *MethodsX*.

In line with the scope of *MethodsX*, this manuscript describes a reproducible method rather than a single set of results. We present RASE-PE, a three-level normalization pipeline that uses Prompt Engineering alone—without fine-tuning—to convert Architecture, Engineering and Construction (AEC) technical standards into the computable RASE format (Requirement, Applicability, Selection, Exception) required for automated BIM compliance checking. The method decomposes the conversion into three successive levels: (N1) atomic segmentation of normative sentences, (N2) identification of RASE operators, and (N3) semantic JSON structuring.

We believe the method fits *MethodsX* for the following reasons:

- It is a transferable, step-by-step procedure that other researchers and practitioners can reproduce and adapt to their own standards and models.
- It was validated on a real dataset of 79 annotated NBR 9050 standards across five experiments, using thirteen metrics from five families (lexical, semantic, distance-based, generation-oriented, and classification), reaching semantic similarity close to 0.90 in the best scenarios.
- It relies exclusively on open-source LLMs (Llama, Dolphin, Gemma, Mistral, Alpaca, and Qwen) served locally through Ollama, under an identical dataset, prompt, and hardware protocol, so the full procedure can be reproduced without proprietary tools.

The manuscript follows the *MethodsX* article structure, includes a CRediT author statement and a data availability statement, and reports the method in sufficient detail to be repeated.

We confirm that this work is original, has not been published previously, and is not under consideration by any other journal. All authors have approved the submission. The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

Thank you for considering our submission.

Sincerely,

Eike Natan Sousa Brito (on behalf of all authors)
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